

ARIN5102 2025-26 Fall Semester Project

This project is on your use and understanding of GPT-4o. While not one of the best LLMs out there anymore, it is still one of the better ones that are available on `genai.ust.hk`, which can also be accessed through API on azure.

Problem 1. (20%) *Due by Thursday, November 20 on canvas.*

Provide 5 examples of problematic behaviours by GPT-4o. These could be incorrect or inappropriate answers to your questions, or correct answers but incorrect reasoning behind the answers. You should be able to find some examples on the internet. You are allowed to re-use them. However, if you do, you must refer to the source. Many of them are math-related, like counting the number of characters in a sentence. Try to be innovative. For example, can you come up with an example that shows GPT-4o simply did not understand your questions?

Each of your examples must be accompanied by the answers from GPT-4o as provided by our `genai.ust.hk`, either through browser or through API, and your analyses of why the answers are problematic.

Problem 2. (80%) Modified Turing Test game competition.

A modified Turing Test game has three participants: a human answerer (H), a machine controlled by (H) through a prompt (M, GPT-4o), and a human questioner (Q). Before the game, (H) composes a prompt for M to use to answer the questions from (Q). Naturally, this prompt is unknown to (Q). For the game, (Q) asks the same question to both (H) and (M), and after receiving the answers from both (H) and (M), decides which one was from (H). If (Q)'s choice is correct, then (Q) receives 1 point while (H) gets -1 point. Otherwise, (Q) gets -1 point while (H) receives 1 point. So this is a zero-sum game between the two humans (H) and (Q).

The prompt by (H) for (M) must not always output the same answer like "I don't know". It can ask GPT-4o to do role playing such as: "pretend you are a high school student with good knowledge in English. If you think such a person can not reasonably answer the given question, then reply with 'I don't know'." It's in (H)'s interest to have (M) output an answer that is not too long and tedious. Otherwise, (Q) can then easily guess which one is from (M). There will be time limits, so (Q)'s question cannot be too complex.

We will conduct a practice run first. We may adjust the rules for the competition based on the result and the feedbacks that we receive after the practice run. The practice run has 0 point but is mandatory: your score in it will not count but your final score for the competition will be deducted by 10% if you do not participate in the practice run without the prior approval from the teaching team.

Practice Run

- Each student will participate in 20 matches, 10 as a human answerer (H) and 10 as a human questioner (Q). The pairing of players will be random: when you play (H), 10 random students will be selected to play (Q) against you, and when you play (Q),

another 10 random students will be selected to play (H) against you.

- In each match, as an (H), you will see all three questions Q1, Q2, Q3 from the same (Q) at the same time. You can answer them anyway you want but remember that you want (Q) to make a mistake so you may want to imitate how (M) will answer them given your prompt: (M)'s three answers A1, A2, A3 to (Q)'s three questions will be generated using GPT-4o as follows: A1 will be the output from (M) given: Your prompt + Q1; A2 will be the output from (M) given: Your prompt + Q1 + A1 + Q2; A3 will be the output from (M) given: Your prompt + Q1 + A1 + Q2 + A2 + Q3.
- In each match, as an (Q), you will see three pairs of answers to your three questions, and for each pair, you select which one is from (H).
- So in each match, you can get the maximum 3 points, and the minimum -3 points, whichever role you played.
- Submit your prompt (when you play (H)) and three questions to ask (when you play (Q)) on canvas by **Thursday, October 16**. Your prompt can be as long as you want, but obviously should not be longer than the maximum number of tokens that can be accepted by GPT-4o. You should also be aware that long prompts may not work well. Each of your questions is limited to 400 characters including spaces.
- Practice run: in class, during the second half of the lecture time on **Thursday, October 23**. You will be given 1 hour to answer 30 questions from the 10 matches you will participate as (H). After a short break, you will have 20 minutes to play the 10 matches that you act as (Q).

Competition (Subject to change depending on how the practice run went.)

- Submit your prompt (when you play (H)) and three questions to ask (when you play (Q)) on canvas by **Thursday, November 6**.
- Competition day: in class, during the second half of the lecture time on **Thursday, November 13**. You will be given 1 hour to answer 30 questions from the 10 matches you will participate as (H). After a short break, you will have 20 minutes to play the 10 matches that you act as (Q).

Your score for this question will be proportional to the total points that you won in the 20 matches during the competition. Below is a tentative marking scheme:

- The full mark for this question is 80.
- You get 0 if you did not participate in the competition without prior approval from the teaching team.

- Every point is worth 0.5 mark, so if you get 30 points, then you get the full mark 80. If you get -30 points, then you get the partial mark 50, which is the base mark for participating in the competition.
- Your total mark will be deducted by 10% if you do not participate in the practice run without prior approval from the teaching team.